# A Physics-guided Hamiltonian Neural Network Framework for Reduced Magnetic Catheter Dynamics

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**Abstract**

This paper develops a physics-guided Hamiltonian neural network (HNN) framework for reduced magnetic catheter dynamics through continuum reduction, Hamiltonian learning and local KAM analysis. A rod-mechanics formulation motivates a Galerkin reduction with conservative and nonconservative terms. The resulting finite-dimensional model is written as a canonical Hamiltonian system for its conservative part and as a port-Hamiltonian system when damping, fluid forces, contact losses or control inputs are present. We then define real and complex internal parameterizations of a Hamiltonian neural network and separate a fixed quadratic physical prior from the representation choice. The retained experiment reports 96 COMSOL trajectories, split by complete trajectories into 64 training, 16 validation and 16 test trajectories, with five training seeds for each model. An independent explicit conservative quartic polynomial is used to compare a central fourth-order generating-function scheme, implicit midpoint, classical RK4 and two-stage fourth-order Gauss–Legendre integration. The fixed prior improves finite-time prediction under both real and complex parameterizations, while the available design does not establish a universal real/complex advantage. The complex-coordinate formulation makes explicit that complex hidden features do not by themselves represent physical dissipation. Finally, a quartic normal-form argument gives a conditional qualitative local KAM family for the explicit conservative polynomial at sufficiently small amplitudes. The conclusion is deliberately object-specific: a nonzero training loss can be interpreted as a Hamiltonian perturbation only under a common canonical chart and a suitable derivative error bound, and the KAM result does not automatically transfer to the learned or material system.

**Keywords**

Hamiltonian neural networks, complex-valued neural networks, magnetic flexible catheters, Galerkin reduction, port-Hamiltonian systems, symplectic integration, KAM theory

## 1 Introduction

Flexible magnetic catheters are distributed mechanical systems. Their state is affected by elastic strain, inertia, magnetic actuation, fluid interaction, boundary contact and, in some settings, gravity. A spatial rod model is therefore a physically meaningful starting point (Antman 2005; Simo 1985; Abbott et al. 2020), but a model that is useful for prediction must also specify a finite-dimensional state, an energy convention, the treatment of nonconservative ports and a time-stepping rule.

The modelling argument proceeds as follows: coupled continuum equations are reduced by a Galerkin basis, the conservative core is expressed in Hamiltonian form, a Hamiltonian neural network learns the unknown energy correction, and local invariant-torus structure is examined through KAM theory. The central scientific requirement is that each statement be attached to the object for which it is valid. The COMSOL reference trajectories represent a material and forcing configuration. The four HNNs used in the learning experiment close their explicit ports and generate a vector field from a real scalar output. The KAM calculation is performed on a separately specified analytic conservative polynomial. These objects motivate one another, but they are not identified without additional modelling and error estimates.

Two distinctions are essential. First, a fixed quadratic energy prior and a real/complex neural representation are

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different factors and should be compared separately. Second, a complex internal representation is not automatically a complex physical energy: in the main learning experiment the network still returns a real scalar Hamiltonian. Dissipation and control are treated through port variables. This avoids interpreting an imaginary network feature as a material loss potential, while retaining the complex-coordinate derivation that motivates the original idea.

The paper makes four contributions. (i) It supplies a mechanically grounded reduction from a coupled rod model to a finite-dimensional Hamiltonian or port-Hamiltonian system. (ii) It gives a consistent real/complex CHNN formulation, including the Wirtinger derivative factors for the coordinate convention used here. (iii) It reports a two-by-two prior/representation comparison and an independent integration study without attributing one result to the wrong factor. (iv) It states the KAM conclusion conditionally for the explicit conservative polynomial and explains what additional evidence would be needed to transfer it to a learned or material system.

## 2 Coupled continuum mechanics and Galerkin reduction

### 2.1 Kinematics, forces and moments

Let $r(s,t)$ be the centreline and $D = (d_1, d_2, d_3) \in SO(3)$ the director frame, for $s \in [0, L]$. Define body strain and curvature by

$$\Gamma = D^{\mathsf{T}} r_s, \qquad D_s = D\widehat{\kappa}, \quad \widehat{\kappa} v = \kappa \times v. \tag{1}$$

A shearable Cosserat model retains all three components of $\Gamma$. In the unshearable Kirchhoff specialization, $\Gamma_1 = \Gamma_2 = 0$ and $r_s = (1+\epsilon)d_3$. Inextensibility imposes the additional constraint $\epsilon = 0$.

To avoid mixing spatial and body components, write the balance laws in spatial form. Let $n$ and $m$ be the internal force and moment, $\ell$ the angular momentum per unit reference length, and $\rho A$ the reference linear density. Then

$$\begin{aligned} \rho A r_{tt} &= n_s + f_{\mathrm{mag}} + f_{\mathrm{hydro}} + f_{\mathrm{contact}} + f_g, \\ \ell_t &= m_s + r_s \times n + \tau_{\mathrm{ext}}. \end{aligned} \tag{2}$$

Here $\tau_{\mathrm{ext}}$ includes magnetic and contact couples. If $\omega_b$ and $\mathcal{J}_b$ are body angular velocity and body inertia per reference length, then $\ell = D\mathcal{J}_b\omega_b$. Consequently $\ell_t = D(\mathcal{J}_b\dot{\omega}_b + \omega_b \times \mathcal{J}_b\omega_b)$ for constant $\mathcal{J}_b$; its right-hand side must also be transformed by $D^{\mathsf{T}}$ when body equations are used.

For a quadratic strain energy density, a shearable constitutive model is

$$D^{\mathsf{T}} n = K_\Gamma(\Gamma - \Gamma_0), \qquad D^{\mathsf{T}} m = K_\kappa(\kappa - \kappa_0). \tag{3}$$

In the constrained Kirchhoff limit, transverse forces are constraint reactions, not the values of a shear law at zero shear. The constrained weak form supplies those reactions.

For a magnetic dipole with body-fixed moment $\mu_b$, put $\mu = D\mu_b$. At fixed orientation the dipole model gives

$$\begin{aligned} U_{\mathrm{mag}} &= -\mu \cdot B(r,t), \\ f_{\mathrm{mag}} &= \nabla_r(\mu \cdot B), \qquad \tau_{\mathrm{mag}} = \mu \times B. \end{aligned} \tag{4}$$

A distributed model integrates dipole energy along the magnetized portion. The moment may rotate with the rod; the spatial derivative in the force formula holds that orientation fixed. A Maxwell-stress formulation is an alternative description of the same interaction and requires its own control surface (Abbott et al. 2020).

Hydrodynamic loading is retained as a general force. A drag term can depend on the relative velocity $v_f - r_t$, while acceleration-dependent added mass must either be moved consistently into the inertial operator or kept on the force side, once only. No particular fluid coefficients are inferred from the trajectory summaries.

For stationary-wall contact, let $\delta$ increase on penetration and let $n_c$ point into the wall so that $\dot{\delta} = n_c \cdot r_t$ locally. An illustrative nonadhesive penalty law is

$$\begin{aligned} f_{\mathrm{contact}} &= -\chi_c(k_c\delta^n + c_c[\dot{\delta}]_+)n_c + f_{\mathrm{friction}}, \\ \chi_c &= \mathbf{1}_{\{\delta>0\}}, \qquad [a]_+ = \max(a, 0). \end{aligned} \tag{5}$$

The normal damper is active only during penetration and compression. This nonsmooth contact model belongs to the mechanical description, not to the analytic polynomial used for KAM theory.

### 2.2 Variational reduction and the four-mode specialization

A finite-dimensional configuration map $\mathcal{Q}(q)$ includes both the centreline and the director field and respects essential boundary conditions and director orthogonality. Its centreline component can take the form

$$r(s,q) = r_0(s) + \Phi(s)q. \tag{6}$$

A centreline expansion alone does not specify rotational inertia. With a consistent director map, the kinetic energy is

$$T = \frac{1}{2}\int_0^L \left(\rho A|r_t|^2 + \omega_b^{\mathsf{T}}\mathcal{J}_b\omega_b\right)\mathrm{d}s = \frac{1}{2}\dot{q}^{\mathsf{T}}M(q)\dot{q}. \tag{7}$$

For time-independent supports and configuration maps, the reduced Lagrange–d'Alembert equations are

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + \nabla V(q) = Q_{\mathrm{nc}}(q,\dot{q},t), \tag{8}$$

where $C\dot{q}$ denotes the inertial terms generated by this kinetic energy. Time-dependent supports or potentials require the corresponding additional terms. Generalized forces follow from virtual work, including rotational and boundary work:

$$Q_i = \int_0^L \left(f\cdot\partial_{q_i}r + \tau\cdot\eta_i\right)\mathrm{d}s + Q_i^{\partial}, \tag{9}$$

where $\delta\vartheta = \sum_i \eta_i\delta q_i$ is the spatial virtual rotation. A potential force is included in $V$ or in $Q$, never both.

These equations provide the mechanical motivation for reduction. They do not reconstruct the unprovided COMSOL configuration, modal basis or boundary data. The experiment retained from the source manuscript uses four modes and a constant mass matrix $M_4$, with

$$p = M_4\dot{q}. \tag{10}$$

Constant modal mass follows, for example, from a fixed linear displacement basis and a fixed kinetic metric. A general nonlinear director parametrization can instead give $M(q)$. The experiment retains its reported $M_4$; its adequacy for the full material system remains a modelling question.

## 3 Hamiltonian and port-Hamiltonian formulation

### 3.1 Legendre transform and canonical equations

For the general reduced conservative part define

$$p = M(q)\dot{q}, \qquad H(q,p) = \frac{1}{2}p^{\mathsf{T}}M(q)^{-1}p + V(q). \tag{11}$$

Let $z = (q,p) \in \mathbb{R}^{2n}$ and

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \qquad \Omega = \sum_{j=1}^{n}\mathrm{d}q_j \wedge \mathrm{d}p_j. \tag{12}$$

The closed conservative system is

$$\dot{z} = J\nabla H(z), \qquad \dot{q} = \nabla_p H, \qquad \dot{p} = -\nabla_q H. \tag{13}$$

If $N(q) = M(q)^{-1}$ and $H = \frac{1}{2}p^{\mathsf{T}}N(q)p + V(q)$, then

$$\dot{q} = N(q)p, \qquad \dot{p}_i = -\frac{1}{2}p^{\mathsf{T}}N_{,q_i}(q)p - V_{,q_i}(q). \tag{14}$$

The first term in $\dot{p}_i$ is the geometric contribution of the kinetic metric. It disappears only when the inverse mass matrix is constant. Likewise, a nonlinear potential requires its true gradient; $K(q)q$ is only a special linear or quadratic case.

### 3.2 Nonconservative forces as ports

For general reduced loads, the canonical equation is

$$\dot{z} = J\nabla H(z) + \begin{pmatrix} 0 \\ Q_{\mathrm{nc}} \end{pmatrix}. \tag{15}$$

If the load admits $Q_{\mathrm{nc}} = -D_d\nabla_p H + B_u u$ with $D_d = D_d^{\mathsf{T}} \succeq 0$, it has the port-Hamiltonian form (Zhong et al. 2020)

$$\dot{z} = (J-R)\nabla H + Gu, \qquad y_{\mathrm{port}} = G^{\mathsf{T}}\nabla H, \qquad R = \mathrm{diag}(0, D_d), \qquad G = \begin{pmatrix} 0 \\ B_u \end{pmatrix}. \tag{16}$$

Consequently

$$\dot{H} = -\nabla H^{\mathsf{T}}R\nabla H + u^{\mathsf{T}}y_{\mathrm{port}} + \partial_t H. \tag{17}$$

Forces without this decomposition remain explicit generalized forces or require additional state variables. Contact switching and moving-fluid energy injection are not automatically represented by a positive-semidefinite dissipation matrix. Standard conservative KAM theory concerns the smooth autonomous core with the nonconservative ports closed.

### 3.3 Four-mode learning coordinates

For the fixed four-mode matrix, let $U$ satisfy, up to stored numerical precision,

$$U^{\mathsf{T}}M_4U = I, \qquad U^{\mathsf{T}}K_4U = \Lambda^2. \tag{18}$$

Define the reconstruction matrix by

$$S = \mathrm{diag}(S_q, S_p), \qquad S_q = U\Lambda^{-1/2}, \qquad S_p = M_4U\Lambda^{1/2}. \tag{19}$$

Since $S_q^{\mathsf{T}} S_p = I$, $S^{\mathsf{T}} J S = J$. Its direction is

$$x = (q, p) = Sy, \qquad y = (Q, P) = S^{-1}x. \tag{20}$$

In exact balancing, substitution into the physical quadratic energy gives

$$\begin{aligned} H_2(y) &= \tfrac{1}{2} P^{\mathsf{T}} M_{\mathrm{norm}}^{-1} P + \tfrac{1}{2} Q^{\mathsf{T}} K_{\mathrm{norm}} Q, \\ M_{\mathrm{norm}}^{-1} &= \Lambda, \qquad K_{\mathrm{norm}} = \Lambda. \end{aligned} \tag{21}$$

Thus the balanced mass is $\Lambda^{-1}$, not the identity. If $H_x(x) = H_y(S^{-1}x)$, then

$$\begin{aligned} \nabla_x H_x &= S^{-\mathsf{T}} \nabla_y H_y, \\ \dot{x} &= S J \nabla_y H_y = J \nabla_x H_x. \end{aligned} \tag{22}$$

Automatic differentiation acts on $y$. Training scales used in the loss are weights on errors and are not another canonical coordinate map. Time remains measured in seconds; this transformation does not supply an unverified physical calibration of the learned energy.

# 4 Complex coordinates and Hamiltonian neural networks

## 4.1 Complex canonical coordinates

For paired canonical coordinates $y = (Q, P)$ define

$$\xi = \frac{Q + iP}{2}, \qquad \bar{\xi} = \frac{Q - iP}{2}. \tag{23}$$

Then $Q = \xi + \bar{\xi}$ and $P = -i(\xi - \bar{\xi})$. The canonical Poisson bracket is

$$\{F, G\} = \nabla_Q F^{\mathsf{T}} \nabla_P G - \nabla_P F^{\mathsf{T}} \nabla_Q G, \tag{24}$$

and the coordinate convention gives

$$\{\xi_j, \bar{\xi}_k\} = -\frac{i}{2} \delta_{jk}. \tag{25}$$

For a real Hamiltonian $H(Q, P)$, Wirtinger differentiation yields

$$\dot{\xi} = -\frac{i}{2} \frac{\partial H}{\partial \bar{\xi}}, \qquad \dot{\bar{\xi}} = \frac{i}{2} \frac{\partial H}{\partial \xi}. \tag{26}$$

The factors in (26) depend on the definition in (23); changing the factor in the coordinate definition changes the bracket and the evolution coefficient.

A formal complexification may be written as $\mathcal{H} = H + iG$. Under the convention

$$\dot{\xi} = -\frac{i}{2} \frac{\partial \mathcal{H}}{\partial \bar{\xi}}, \tag{27}$$

the additional term is $\frac{1}{2} \partial_{\bar{\xi}} G$. In real coordinates this gives $\dot{Q} = H_P + G_Q$ and $\dot{P} = -H_Q + G_P$, with energy contribution $\nabla H \cdot \nabla G$ of unrestricted sign. This term is not automatically a physical damping force. To represent a prescribed port, $G$ must satisfy the corresponding integrability conditions in the chosen coordinates. For this reason the main experiment uses a real output Hamiltonian and keeps physical nonconservative effects in (16); the complex variables are used to describe an internal representation and to derive the conservative complex-coordinate equations.

## 4.2 Real and complex CHNNs

The no-prior model learns a scalar energy $H_\theta(y)$ and generates

$$\dot{y}_\theta = J \nabla_y H_\theta(y). \tag{28}$$

The gray-box model adds the fixed prior,

$$H_\theta^{\mathrm{prior}}(y) = H_2(y) + H_{\mathrm{res},\theta}(y). \tag{29}$$

The real network uses real affine maps and real hidden features. The recorded complex implementation instead uses $z_{\mathrm{net}} = (Q + iP)/\sqrt{2} = \sqrt{2}\xi$ as its input, followed by complex affine maps and separate real and imaginary hyperbolic tangents,

$$\sigma(u + iv) = \tanh(u) + i \tanh(v), \tag{30}$$

and a real-part readout. The recorded widths are $4 \to 45 \to 45 \to 45 \to 1$ for the complex network and $8 \to 64 \to 64 \to 64 \to 1$ for the real network. The reported learning experiment advances $J \nabla H_\theta$ with RK4; it does not train or use the central generator derived below. Its learned object is still a real scalar energy (Greydanus et al. 2019; Trabelsi et al. 2018). Split complex activations need not satisfy Cauchy–Riemann equations; holomorphicity is required only if a holomorphic model is explicitly intended.

The four factorial models are $R$ (real/no prior), $GR$ (real/gray-box), $C$ (complex/no prior) and $GC$ (complex/gray-box). A residual MLP is a separate descriptive baseline. The prior effects are $R - GR$ and $C - GC$, representation effects are $R - C$ and $GR - GC$, and the interaction is

$$(GR - GC) - (R - C). \tag{31}$$

## 4.3 Representation and approximation scope

A split-complex network with a real-part readout can contain a real tanh network of the same hidden widths. For a

real first layer $A_Q Q + A_P P + b$ and input $z_{\rm net} = (Q + iP)/\sqrt{2}$, choose the complex first-layer weight $\sqrt{2}(A_Q - iA_P)$ and real bias $b$. Its real preactivation is the required real affine map. Choose subsequent weights and biases real; the imaginary channel then does not enter the real readout. This construction does not embed the reported width-64 real network in the reported width-45 complex network.

For growing widths and a target with a $C^m$ extension to a neighbourhood of a compact set, smooth tanh approximation gives finite-order derivative approximation (Hornik 1991). Extra hidden layers can approximate the identity on bounded intermediate ranges using $\tanh(\delta v)/\delta$ as $\delta \to 0$. The real-part embedding transfers this existence statement to the growing split-complex family. It proves neither an optimization guarantee nor an advantage at a fixed parameter budget, and gives no analytic-strip error certificate.

### 4.4 Training loss and dynamical perturbation

Let $D_{\rm train} = \mathrm{diag}(d_1, \ldots, d_8)$ be fixed from the training data. For a trajectory with 801 saved states, the stored rollout metric is

$$X_{msi} = \frac{1}{800} \sum_{k=1}^{800} \frac{1}{8} \left\| D_{\rm train}^{-1} (y_{bmsi,k} - y_{i,k}) \right\|_2^2 . \tag{32}$$

This metric is a finite-time average and is not a uniform vector-field norm.

Suppose, conditionally, that a learned Hamiltonian and a target Hamiltonian are defined in the same canonical chart and with the same energy gauge. Let $\Delta H_\theta = H_\theta - H_*$. Then

$$\dot{y}_\theta - \dot{y}_* = J \nabla \Delta H_\theta . \tag{33}$$

A sufficient local perturbation statement requires a derivative bound such as

$$\sup_{y \in \mathcal{U}} \| \nabla \Delta H_\theta(y) \| \le \delta . \tag{34}$$

A nonzero rollout loss, by itself, does not establish (34). This only controls the vector field in a real-domain supremum norm. It does not suffice for analytic KAM: that theorem requires smallness on a common complex domain, or a separate finite-regularity KAM theorem with its own derivative and smallness conditions. The learning-to-KAM route of Chen et al. (2022) motivates this distinction; its hypotheses have not been verified for the present rollout metric.

## 5 Normal form and conditional local KAM structure

### 5.1 Conservative polynomial and coordinate convention

To separate numerical structure from learning, define the explicit real polynomial

$$H_{\rm dec}(W) = \frac{1}{2} \sum_{j=1}^{4} \omega_j (X_j^2 + Y_j^2) \tag{35}$$

$$+ \sum_{a \le b \le c} c_{abc} W_a W_b W_c \tag{36}$$

$$+ \sum_{a \le b \le c \le d} c_{abcd} W_a W_b W_c W_d, \quad W = (X, Y). \tag{37}$$

The indices range from 1 to 8 and each sorted monomial is counted once. The four frequencies, interpreted as exact decimal rationals for this polynomial, are

$$\omega_1 = 34.95334005159986, \quad \omega_2 = 35.22520200193075,$$
$$\omega_3 = 217.78628967842488, \quad \omega_4 = 220.49013432955627. \tag{38}$$

The cubic and quartic coefficients are those specified by the original polynomial record. That record is not among the supplied files. Accordingly, the symbolic argument below is complete for a polynomial with those coefficients and the stated twist, but its application to the recorded polynomial remains subject to recovery of the coefficient record. The polynomial is independent of the learned energy and is not identified with the full material Hamiltonian.

### 5.2 Normal form of the explicit polynomial

The KAM object is the explicit polynomial $H_{\rm dec}$ in its own canonical modal coordinates $W = (X, Y)$. The CHNN convention in Section 4.1 is not reused here. Define

$$a_j = \frac{X_j + iY_j}{\sqrt{2}}, \qquad I_j = a_j \bar{a}_j, \qquad \{a_j, \bar{a}_k\} = -i\delta_{jk}. \tag{39}$$

Then

$$H_{2,\rm dec} = \sum_{j=1}^{4} \omega_j I_j = \omega \cdot I. \tag{40}$$

Write $H_{\rm dec} = H_2 + H_3 + H_4$ and let $L_\chi F = \{\chi, F\}$. On $a^\mu \bar{a}^\nu$,

$$\{a^\mu \bar{a}^\nu, H_2\} = -i d_{\mu\nu} a^\mu \bar{a}^\nu, \qquad d_{\mu\nu} = (\mu - \nu) \cdot \omega. \tag{41}$$

Treating the displayed decimal frequencies as the exact coefficients of the specified polynomial, enumeration gives

$$\min_{|\mu|+|\nu|=3} |d_{\mu\nu}| = 32.24949540046847, \tag{42}$$

$$\min_{|\mu|+|\nu|=4,\, d\neq 0} |d_{\mu\nu}| = 0.27186195033089. \tag{43}$$

The cubic kernel is empty; the quartic kernel consists exactly of $\mu = \nu$, hence of the ten action monomials. These finite checks do not assert a Diophantine inequality for all integer vectors.

Let $\Pi$ project quartic terms onto this kernel and set kernel parts of the generators to zero. Coefficientwise,

$$(\chi_3)_{\mu\nu} = -i(H_3)_{\mu\nu}/d_{\mu\nu}, \tag{44}$$

$$G_4 = H_4 + \tfrac{1}{2}\{\chi_3, H_3\}, \qquad \widehat{Z}_4 = \Pi G_4, \tag{45}$$

$$(\chi_4)_{\mu\nu} = -i(G_4)_{\mu\nu}/d_{\mu\nu} \quad (d_{\mu\nu} \neq 0), \qquad \Pi\chi_4 = 0. \tag{46}$$

The finite Lie calculation is

$$\begin{aligned}
[e^{L_{\chi_4}} e^{L_{\chi_3}} H_{\mathrm{dec}}]_{\leq 4} &= H_2 + H_3 + \{\chi_3, H_2\} && (47)\\
&\quad + H_4 + \{\chi_3, H_3\} && (48)\\
&\quad + \tfrac{1}{2}\{\chi_3, \{\chi_3, H_2\}\} && (49)\\
&\quad + \{\chi_4, H_2\} && (50)\\
&= H_2 + \widehat{Z}_4. && (51)
\end{aligned}$$

The definitions solve the homological equations exactly: $\{\chi_3, H_2\} = -H_3$ and $\{\chi_4, H_2\} = -(G_4 - \Pi G_4)$. This proves the cancellation algebraically. Real input coefficients have conjugate symmetry in $(a, \bar{a})$; the generators inherit that symmetry and define real Hamiltonians. A calculation using rounded generators must retain its cubic residual $E_3 = H_3 + \{\chi_3, H_2\}$, including the quartic correction $\frac{1}{2}\{\chi_3, E_3\}$. A small residual is not an exact cancellation certificate. Since $X_{\chi_3} = O(\|W\|^2)$ and $X_{\chi_4} = O(\|W\|^3)$, the maps are analytic and symplectic on a sufficiently small complex neighbourhood. With

$$C = \phi_{\chi_3}^{-1} \circ \phi_{\chi_4}^{-1}, \tag{52}$$

where $\phi_\chi$ is the time-one Hamiltonian flow, one obtains

$$H_{\mathrm{dec}} \circ C = h_{\mathrm{NF}}(I) + R_{\mathrm{Lie}}(W), \quad R_{\mathrm{Lie}} = O(\|W\|^5), \tag{53}$$

$$h_{\mathrm{NF}}(I) = \omega \cdot I + \tfrac{1}{2} I^{\mathsf{T}} T_b I. \tag{54}$$

The remainder is the full analytic remainder after this finite transformation, not an assumed material-model error.

**Table 1.** Displayed quartic action coefficients used to report the twist matrix. These decimal values are not a substitute for the missing exact coefficient file.

| $\beta$ | $c_\beta$ |
|---|---|
| $(2,0,0,0)$ | 434470615578.06039327 |
| $(1,1,0,0)$ | 583279003999.93543432 |
| $(1,0,1,0)$ | 3734689434505.0453818 |
| $(1,0,0,1)$ | −4585094327803.5669993 |
| $(0,2,0,0)$ | 440827566852.01874934 |
| $(0,1,1,0)$ | −2375967766950.0738049 |
| $(0,1,0,1)$ | 12143155842211.582250 |
| $(0,0,2,0)$ | 5306062760124380 0.310 |
| $(0,0,1,1)$ | −1633954193532226930.8 |
| $(0,0,0,2)$ | $-1.6345060222020102537 \times 10^{20}$ |

### 5.3 Twist and qualitative persistence

The original analysis record reports the following determinant and inverse bound for its exact normal form:

$$\det T_b \simeq -1.5902519571118942207 \times 10^{61}, \tag{55}$$

$$\|T_b^{-1}\|_\infty \leq 3.44184331720408 \times 10^{-12}. \tag{56}$$

Table 1 retains the ten displayed coefficients of $\widehat{Z}_4 = \sum_{|\beta|=2} c_\beta I^\beta$. Here $(T_b)_{ii} = 2c_{2e_i}$ and $(T_b)_{ij} = c_{e_i+e_j}$ for $i \neq j$. Reconstructing a matrix from these decimals reproduces the reported determinant and inverse norm to the displayed precision. It does not verify the link from the missing $H_3, H_4$ coefficients to that matrix, nor replace the original exact inverse bound. The persistence result below explicitly assumes that link and nondegeneracy.

The integrable truncation has

$$X_j = \sqrt{2I_j}\cos\phi_j, \qquad Y_j = -\sqrt{2I_j}\sin\phi_j, \tag{57}$$

$$\dot{\phi} = \omega + T_b I. \tag{58}$$

It has invariant action tori even at resonance; KAM concerns persistence after $R_{\mathrm{Lie}}$ is restored.

**Proposition 1.** Conditional local KAM family. *Let $H_{\mathrm{dec}}$ be a real quartic Hamiltonian with quadratic frequencies* (38)*. Suppose its exact quartic normal form has $\det T_b \neq 0$. Then sufficiently small scaled copies of a fixed positive-action shell contain a positive-measure union of analytic Lagrangian invariant four-tori for the exact flow. This gives neither a numerical amplitude threshold nor a conclusion for a learned or material Hamiltonian.*

**Proof.** The finite divisor checks and homological construction give the analytic form (54). Fix a bounded action domain $D$ whose closure lies in $(0,\infty)^4$, and an inner domain with positive boundary margin. A common complex neighbourhood of $D$, together with an angle strip of width $s > 0$,

permits one analytic branch of each square root in the action–angle map.

Set $W = \varepsilon w$, $I = \varepsilon^2 A$ and $\sigma = \varepsilon^2 t$. The Hamiltonian for the rescaled time is

$$\begin{aligned} K_\varepsilon &= \varepsilon^{-4}(H_{\mathrm{dec}} \circ C)(\varepsilon w) \\ &= b_\varepsilon \cdot A + \tfrac{1}{2} A^{\mathsf{T}} T_b A + P_\varepsilon(A, \phi), \\ b_\varepsilon &= \varepsilon^{-2}\omega, \qquad \|P_\varepsilon\| \le C_* \varepsilon. \end{aligned} \tag{59}$$

The supremum norm is on the common complex domain, and $C_*$ is independent of sufficiently small $\varepsilon$. The estimate follows from the full analytic $O(\|W\|^5)$ remainder, not a truncation at degrees five and six.

The frequency domain is $\Omega_\varepsilon = b_\varepsilon + T_b D$. On a fixed inner part of this translated domain the slab $|k \cdot \lambda| < \alpha |k|_1^{-4}$ has volume at most $C\alpha |k|_1^{-5}$. The constant is independent of translation, because the domain has a fixed shape and bounded cross-sections. Summation over $k \in \mathbb{Z}^4 \setminus \{0\}$ converges. Thus a fixed small $\alpha > 0$ retains a positive-measure Diophantine set without assuming that the equilibrium frequency $\omega$ is itself Diophantine.

For a frequency parameter $\lambda$ define

$$a_\varepsilon(\lambda) = T_b^{-1}(\lambda - b_\varepsilon), \qquad A = a_\varepsilon(\lambda) + u. \tag{60}$$

The translated Hamiltonian $\mathcal{K}_\lambda = N_\lambda + P_\lambda$ has

$$\begin{aligned} N_\lambda(u) &= e_\varepsilon(\lambda) + \lambda \cdot u, \\ P_\lambda(u, \phi) &= \tfrac{1}{2} u^{\mathsf{T}} T_b u + P_\varepsilon(a_\varepsilon(\lambda) + u, \phi). \end{aligned} \tag{61}$$

Choose complex parameter width $\eta$ and action radius $r_*$ so that $r_* + \|T_b^{-1}\|_\infty \eta$ is less than the action-domain margin. Both widths can be independent of $\varepsilon$. On $|u|_\infty < r_*$,

$$\|P\| \le M r_*^2 + C_* \varepsilon, \qquad M = \tfrac{1}{2} \sum_{i,j} |(T_b)_{ij}|. \tag{62}$$

We apply Theorem A of Pöschel (2001), with the corrected proof of Bounemoura (2024), for dimension four and $\tau = 4$. Its analytic-domain and parameter hypotheses are satisfied by the preceding construction. Its smallness requirements are

$$\begin{aligned} |k \cdot \lambda| &\ge \alpha |k|_1^{-4} \quad (k \in \mathbb{Z}^4 \setminus \{0\}), \\ \|P\| &\le c_* \alpha r_* s^5, \qquad \alpha s^5 \le \eta. \end{aligned} \tag{63}$$

First choose $s, \eta, \alpha$ within the domain and measure restrictions. Next choose $r_* > 0$ small enough that $M r_*^2 \le c_* \alpha r_* s^5/2$. Finally decrease $\varepsilon$ until $C_* \varepsilon \le c_* \alpha r_* s^5/2$. This establishes the full smallness condition, including the quadratic term.

The theorem supplies a parameter correction $\varphi_\varepsilon$ and analytic torus embeddings: for each retained frequency $\lambda$, the torus belongs to $\mathcal{K}_{\varphi_\varepsilon(\lambda)}$. These Hamiltonians are translations of the same $K_\varepsilon$. The theorem's Lipschitz estimates, together with the invertible action–frequency map, give a positive-measure union in phase space. The required closeness can be achieved by further reducing $r_*$ and then $\varepsilon$ in the same order. Translating back, applying $W = \varepsilon w$, and applying $C$ preserve positivity of measure for each $\varepsilon > 0$. The physical-time frequencies are $\varepsilon^2 \lambda$.

This is an application of a classical theorem to the specified polynomial class. To assert it for the recorded numerical polynomial, its recovered coefficients must produce the stated nondegenerate twist. The quadratic learning prior alone has zero twist and cannot replace this nonlinear calculation.

### 5.4 Why transfer to the learned system is conditional

Transfer to $H_\theta$ requires a common canonical chart, a consistent energy gauge, analyticity or a stated finite-regularity theorem, a perturbation bound on the theorem's domain and preservation of the nondegenerate frequency map. A finite trajectory loss does not supply these quantities. For a damped or forced port-Hamiltonian system, conservative KAM is not the applicable theorem without a separate dissipative or forced invariant-manifold analysis.

## 6 Numerical experiments

### 6.1 Trajectory split and evaluation

The reported experiment contains 96 COMSOL trajectories. Complete trajectories, rather than individual time samples, are split into 64 training, 16 validation and 16 test trajectories. Each trajectory has 801 states over 0–0.2 s with $\Delta t = 2.5 \times 10^{-4}$ s. The 16 test trajectories belong to six orbit groups. Five independent training seeds are retained for each model, and the epoch-300 checkpoint is used without selecting on test performance.

The supplied manuscript summaries identify counts and orbit groups but provide no trajectory-level forcing record. In particular, it does not establish whether all 96 trajectories share one magnetic-field history $B(t)$, whether field direction or amplitude varies between groups, or whether the groups encode initial conditions, forcing protocols or both. The four HNNs therefore close their explicit ports and should be interpreted as closed-port reduced predictors over the observed trajectories. Before submission, the forcing

protocol must be documented by trajectory-level metadata containing the initial state, orbit-group label, magnetic-field or control history and any fluid or contact parameters. If these inputs vary independently of the observed state, a predictive model must condition on them or augment its state. Even one common time-dependent $B(t)$ need not produce an autonomous state-only vector field: the same state at different times can have different derivatives. A justified autonomous closure, fixed autonomous forcing, or an explicit time/phase/input model is needed. The existing statistics describe the reported dataset comparison while this identifiability question remains open.

The original descriptive metric takes the median over five seeds for each model–trajectory pair and then summarizes the 16 test values. The resulting values are shown in Table 2.

**Table 2.** Descriptive rollout metric for the five models.

| Model | Mean | Median |
|---|---|---|
| $R$ | $1.66533 \times 10^3$ | $1.37061 \times 10^3$ |
| $GR$ | $1.26587 \times 10^1$ | $9.57033 \times 10^0$ |
| $C$ | $1.66528 \times 10^3$ | $1.37065 \times 10^3$ |
| $GC$ | $1.26640 \times 10^1$ | $9.60739 \times 10^0$ |
| Residual MLP | $2.20953 \times 10^9$ | $2.20953 \times 10^9$ |

The fixed prior reduces the finite-test rollout metric under both parameterizations. This is evidence for the prior in the stated finite design, not evidence that the prior recovers every term of the material energy.

## 6.2 Factorial effects and sensitivity

The five effects are summarized in Table 3. Positive values in the first two rows favour including the prior; positive values in the next two rows favour the complex implementation under the stated metric.

The original analysis uses paired trajectory resampling conditional on the five-seed aggregation. The sensitivity analysis was specified after that experiment and uses six whole orbit groups of sizes $2, 2, 4, 2, 2, 4$. If $Y_{mi} = \operatorname{median}_s X_{msi}$ and $D_i = Y_{Ai} - Y_{Bi}$, its two weighting targets are

$$\mu_T = \frac{1}{16}\sum_i D_i, \qquad \mu_G = \frac{1}{6}\sum_{g=1}^{6}\frac{1}{n_g}\sum_{i\in g} D_i. \tag{64}$$

The new primary analysis uses $\mu_G$, seed medians and joint group–seed resampling. Groups are drawn with replacement, retaining every phase; group draws are shared across models. Seed labels are sampled independently for each model and reused across that model's trajectories. The recorded protocol uses 20,000 replicates, linear percentile endpoints 2.5%/97.5%, and nominal five-effect-adjusted endpoints 0.5%/99.5%. Group-only and seed-only resampling, seed means and leave-one checks are sensitivity calculations. The nominal adjustment relies on the working sampling assumptions; simultaneous population coverage is not established.

Both prior effects retain their improvement direction under the prescribed weighting, aggregation and leave-one checks. The sealed analysis does not resolve an additional complex benefit. The post-experiment group/seed sensitivity calculation gives a negative gray-box real-minus-complex contrast, favouring $GR$ in this design, while the no-prior representation contrast and the interaction remain unresolved. The intervals rely on the stated finite-design resampling assumptions and do not establish population coverage or universal representation superiority.

## 6.3 Central generating-function scheme

For an autonomous real Hamiltonian $\dot{u} = J\nabla H(u)$, write the exact endpoints of one step as $u^+ = \Phi_h(u^-)$ and set

$$m = \frac{1}{2}(u^+ + u^-), \qquad \delta = u^+ - u^-. \tag{65}$$

Locally $\delta = F(m, h)$. Differentiating the identity for the exact symplectic map gives $A^{\mathsf{T}}J + JA = 0$ for $A = D_m F$, because

$$\begin{aligned}(I + A/2)^{\mathsf{T}}J(I + A/2) - (I - A/2)^{\mathsf{T}}J(I - A/2)\\ = A^{\mathsf{T}}J + JA.\end{aligned} \tag{66}$$

Consequently $-JF$ is locally a gradient. There is a central generating function $S_h$ such that

$$\nabla_m S_h = -JF, \qquad F = J\nabla_m S_h. \tag{67}$$

For completeness, the formal recurrence used for the fourth-order construction is

$$\begin{aligned}\partial_h F &= \left(I - \tfrac{1}{2}D_m F\right) f\left(m + \tfrac{1}{2}F\right), & f &= J\nabla H,\\ \partial_h S_h &= H\left(m + \tfrac{1}{2}J\nabla S_h\right), & S_0 &= 0,\\ S_1 &= H,\end{aligned} \tag{68}$$

$$S_{k+1} = \frac{1}{k+1}\left[h^k\right] H\left(m + \tfrac{1}{2}J\sum_{j=1}^{k} h^j \nabla S_j\right). \tag{69}$$

For an autonomous Hamiltonian, $S_2 = S_4 = 0$ and

$$S_3 = \frac{1}{24} f^{\mathsf{T}} H'' f, \tag{70}$$

$$g_h = J\nabla(hS_1 + h^3 S_3) \tag{71}$$

$$= hf + \frac{h^3}{24}\left(f''[f, f] - 2f'^2 f\right). \tag{72}$$

**Table 3.** Factorial effects from the reported analysis. Original estimates use equal trajectory weights; new sensitivity estimates use equal group weights. Both use seed medians; they are different estimands.

| Effect | Original | New | Original 95% interval | New adjusted interval |
|---|---|---|---|---|
| $R-GR$ | 1652.68 | 1438.62 | $[1043.01, 2314.78]$ | $[332.78, 2778.00]$ |
| $C-GC$ | 1652.62 | 1438.58 | $[1037.70, 2311.88]$ | $[332.58, 2777.95]$ |
| $R-C$ | 0.05316 | 0.03448 | $[-0.08880, 0.17862]$ | $[-0.11404, 0.98582]$ |
| $GR-GC$ | $-0.00532$ | $-0.00529$ | $[-0.14163, 0.12850]$ | $[-0.01274, -0.00103]$ |
| Interaction | $-0.05848$ | $-0.03977$ | $[-0.17323, 0.06401]$ | $[-0.99095, 0.10964]$ |

The implemented endpoint is the implicit equation

$$u^+ = u^- + g_h\left(\tfrac{1}{2}(u^+ + u^-)\right). \tag{73}$$

Newton's residual $R(v) = v - u^- - g_h((u^- + v)/2)$ has derivative $I - \frac{1}{2}Dg_h$. If the endpoint fixed-point map sends a closed ball into itself and $\kappa = \frac{1}{2}\sup\|Dg_h\| < 1$ there, the contraction theorem gives a unique root in that ball. Newton convergence also requires a sufficiently close initial guess and the usual local smoothness assumptions; contraction alone does not certify arbitrary Newton iterates. For an exact root, the derivative is

$$D\Phi_h = (I - A/2)^{-1}(I + A/2), \qquad A = Dg_h, \tag{74}$$

which is symplectic because $A^{\mathsf{T}}J + JA = 0$. This is a symplecticity statement for the exact implicit map; a finite residual tolerance does not prove exact symplecticity of a program. The truncation gives local error $O(h^5)$ and fixed-time global error $O(h^4)$ under the required smoothness (here it is sufficient that $H \in C^8$). It preserves the modified generating structure, not the original energy exactly.

The ordinary-power expansion $S_h = \sum_{j\geq 1} h^j S_j$ defines the coefficients used above. Time reversal gives $S_{-h} = -S_h$ up to a fixed energy gauge, hence the even coefficients vanish. Differentiating $f^{\mathsf{T}}H''f$ yields $J\nabla(f^{\mathsf{T}}H''f) = f''[f,f] - 2f'^2 f$. This proves the stated cubic correction; matching the exact central displacement through degree four gives local order five. The symplectic Cayley identity and finite-time stability then give the preceding map property and global order under uniform smoothness on a compact trajectory neighbourhood (Feng et al. 1989; Guo et al. 2001; Hairer et al. 2006).

If an approximate root has residual at most $\tau_h$ in the contraction ball, its distance to the exact root is at most $\tau_h/(1-\kappa)$. Under the same fixed-time stability assumptions, a uniform per-step residual contributes $O(\tau_h/h)$ to trajectory error. A bound on residual values does not bound derivatives of the implemented solver map, so it does not by itself control the symplecticity defect. Evaluating $g_h$ requires third derivatives of $H$ and its Newton matrix requires fourth derivatives.

The comparison applies central fourth order, implicit midpoint, classical RK4 and two-stage fourth-order Gauss–Legendre to the same $H_{\mathrm{dec}}$, initial conditions, reference solution and cost accounting (Feng et al. 1989; Hairer et al. 2006). No learned-network cost is inferred from this polynomial experiment.

## 6.4 Integration protocol and results

For the same $H_{\mathrm{dec}}$, the state error and energy change are measured by

$$e_{\mathrm{state}} = \max_t \frac{\|W_{\mathrm{num}}(t) - W_{\mathrm{ref}}(t)\|_2}{\|W_0\|_2}, \qquad e_H = \max_n \frac{|H_{\mathrm{dec}}(W_n) - H_{\mathrm{dec}}(W_0)|}{H_{2,\mathrm{dec}}(W_0)}. \tag{75}$$

For the integration comparison, define

$$u_1 = (1,1,1,1,0,0,0,0)/2, \tag{76}$$

$$u_2 = (1,2,3,4,4,-3,2,-1)/\sqrt{60}, \tag{77}$$

$$A_1 = (r/2)u_1, \qquad A_2 = (r/2)u_2, \tag{78}$$

$$B_1 = ru_1, \qquad B_2 = ru_2. \tag{79}$$

with $r = 1.4810634592307795 \times 10^{-11}$, solver scale $R = 1.0521287803311451 \times 10^{-10}$, grid $h_j = h_0/2^j$ for $j = 0,\ldots,4$, $h_0 = 0.0017810279035513454$, and final time $T = 2019h_0 = 3.5958953372701665$. These four initial conditions and five step sizes give 80 configurations. The selected amplitude is a solver scale, not a KAM radius. On the fixed step-size grid, midpoint did not attain the three prescribed state-error targets. The central fourth-order method had smaller observed energy changes than RK4, but its integration cost was approximately 2.59–4.02 times that of RK4 at the selected state-accuracy targets. Gauss was faster than the central method at $10^{-4}$ and $10^{-5}$, while the central method was faster on the selected $10^{-3}$ configurations. These results establish an energy–cost trade-off on one polynomial, not uniform performance superiority.

The reference is an independently tightened DOP853 solution; the reported reference and Newton-tolerance checks passed their prescribed criteria, but do not give interval-arithmetic certificates. Gauss uses 16 stage-state

unknowns with residuals in state units. Residual and final correction divided by $R$ must both be at most $10^{-13}$. Cost includes derivative assembly, Newton corrections, linear solves, line searches and native-step energy evaluation. The reported cost is the median of three post-warm-up timings in one common single-thread, float64 environment. Table 4 selects among the five measured step sizes; shared preparation costs an additional 57.0300 s. These timings are observations from the original record and have not been rerun from the supplied package.

**Table 4.** Post-warm-up integration-cost ranges (seconds) at the three state-accuracy targets. Shared preparation cost is excluded. "Not attained" means that none of the five tested step sizes reached the target.

| Method | $10^{-3}$ | $10^{-4}$ | $10^{-5}$ |
|---|---|---|---|
| Central fourth order | 0.705–0.750 | 1.011–1.352 | 1.828–2.057 |
| Implicit midpoint | Not attained | Not attained | Not attained |
| Classical RK4 | 0.185–0.187 | 0.370–0.374 | 0.704–0.751 |
| Two-stage Gauss, order 4 | 0.933–0.990 | 0.933–0.990 | 1.425–1.692 |

A torus of the exact flow remains invariant under exact time sampling. This does not establish a torus for an algorithmic map. Such a conclusion needs an appropriate map theorem, exact symplectic structure, step-frequency nonresonance and controlled embedding and residual estimates (Figueras et al. 2017). The present integration errors do not supply these certificates.

## 7 Discussion and modelling guidance

The continuum derivation shows why the Hamiltonian core is natural, but it also identifies which terms must remain outside a conservative potential. The Galerkin step explains the origin of the reduced coordinates and makes the fixed four-mode mass matrix an explicit experimental specialization rather than an unstated physical identity.

The prior and representation effects should be selected in separate comparisons. A lower error for $GR$ or $GC$ establishes that the prescribed quadratic prior helps the stated finite-time prediction task. It does not prove complete material identification. The real/complex comparison is likewise a comparison of the implemented widths, activations, optimization and data protocol. The results do not support a universal ordering of real and complex neural networks.

The complex Hamiltonian derivation remains useful for the original research direction: it gives a compact coordinate language, exposes the canonical bracket and provides a route to normal-form calculations. Its physical interpretation must remain controlled. A formal imaginary component $G$ is not a damping law until the generated vector field is shown to match the desired port and power balance. The mechanical formulation keeps nonconservative loads explicit; the reported learning experiment uses real scalar energies with those explicit ports closed.

The independent polynomial study should also remain separate from the learned field. The central generating-function scheme is an existing structure-preserving method; its observed energy–cost trade-off is a numerical result for $H_{\mathrm{dec}}$. Applying the same method to $H_\theta$ would require a new cost and error comparison because automatic-differentiation and implicit-solve costs depend on the learned network.

**Table 5.** Scope of the main conclusions.

| Object | Supported conclusion and remaining requirement |
|---|---|
| Reduced material model | Hamiltonian core plus port terms after Galerkin reduction; requires boundary, force and truncation specifications. |
| Learned $H_\theta$ | Finite-design prediction comparison; KAM transfer requires common coordinates and analytic/derivative error control. |
| Explicit $H_{\mathrm{dec}}$ | Conditional local KAM family under normal-form, twist, Diophantine and small-remainder assumptions. |
| Discrete integrator | State, energy and cost trade-offs on $H_{\mathrm{dec}}$; RK4 is not generally exact symplectic. |

## 8 Conclusion

The CHNN–Hamiltonian–KAM route is organized here as a single mechanically grounded workflow for reduced magnetic catheter dynamics. The coupled rod equations provide the continuum origin, Galerkin projection gives the finite-dimensional model, and the Legendre transform produces the Hamiltonian core. Damping, hydrodynamic forces, contact losses and control are retained through port-Hamiltonian terms. Real and complex CHNNs are then compared under a factorial prior/representation design, while an independent conservative polynomial is used for integration and local KAM analysis.

In the supplied finite design, the fixed quadratic prior improves prediction under both parameterizations. The available results do not establish a universal real/complex advantage. The central fourth-order scheme trades smaller observed energy changes for higher cost than RK4 at selected targets, with no consistent advantage over Gauss. The explicit polynomial admits a qualitative local KAM family under the stated analytic, twist and Diophantine assumptions. That conclusion is intentionally not transferred to the learned or material system without a common canonical chart and an appropriate derivative or analytic error estimate.

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### Data availability

The present manuscript package contains the LaTeX source and compiled PDF. A complete submission package still needs to include the decimal coefficient file for $H_{\mathrm{dec}}$, the exact normal-form coefficient file, `SOURCE_INDEX.json`, the statistical CSV outputs, the integration CSV outputs and trajectory-level forcing metadata. Until those files are attached to a versioned repository, the numerical KAM constants and sensitivity tables are documented claims from the current analysis record rather than independently reproducible outputs from this ZIP alone.

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